

Hybrid Quantum-Classical Architectures for Federated Learning: A Simulation-Based Study on Design, Optimization, and Scalability

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Abstract

The convergence of quantum computing and federated learning presents a transformative opportunity for building next-generation distributed AI systems. While fully quantum implementations remain constrained by hardware noise, qubit decoherence, and scalability limitations in the Noisy Intermediate-Scale Quantum (NISQ) era, Hybrid Quantum-Classical (HQC) architectures offer a pragmatic and high-performing alternative. This chapter presents a simulation-based investigation of HQC architectures tailored for federated learning environments, implemented using Qiskit. We design parameterized Variational Quantum Circuits (VQCs) integrated with classical neural network layers to construct hybrid models capable of distributed training across federated nodes. Gradient optimization is performed using the parameter-shift rule, and classical aggregation is achieved through a modified Federated Averaging (FedAvg) protocol adapted for hybrid model parameters. Our simulation results demonstrate measurable performance improvements over classical-only baselines in specific learning tasks, while maintaining federated privacy guarantees. We further analyze the impact of quantum noise on model convergence, evaluate communication overhead across distributed nodes, and discuss practical scalability strategies for heterogeneous quantum-classical environments. This work bridges theoretical foundations with reproducible simulation experiments, providing a practical and accessible reference for researchers exploring quantum-enhanced federated intelligence.

Index Terms

Hybrid Quantum-Classical Systems, Federated Learning, Variational Quantum Circuits, Quantum Noise Mitigation, NISQ, FedAvg, Distributed Intelligence, Quantum Machine Learning